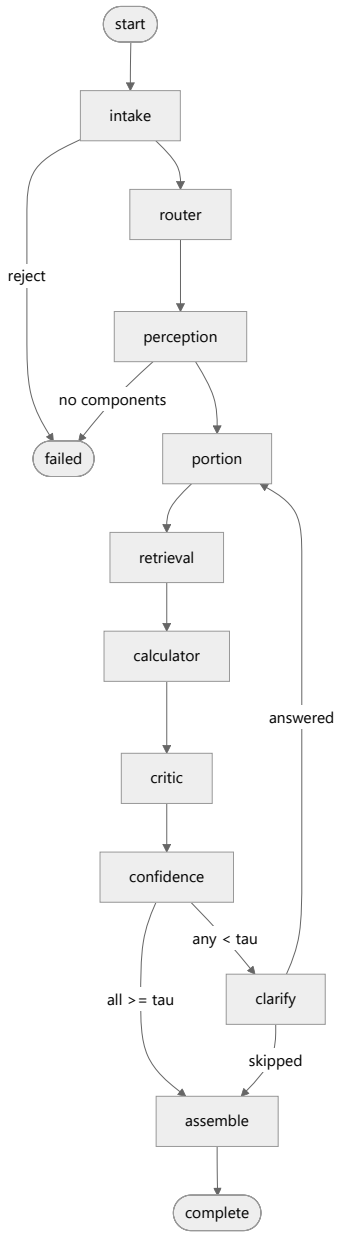
Architecture

Adaptive Food Recognition and Nutritional Estimation Using Multimodal AI — Team The Invincibles, GVP College of Engineering (CSE).

This is the consolidated architecture reference: how the system is shaped, how a request moves through it, what it is built on, and what every stage is contractually allowed to output. Deep detail lives in `docs/` (links at the end).

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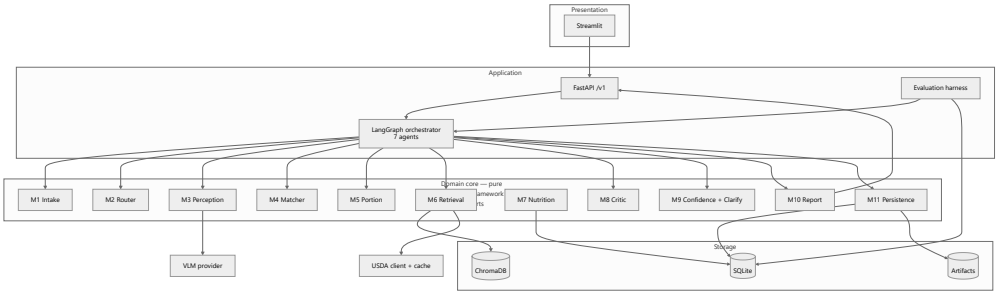
1. Multi-agent graph



2. The problem the architecture solves

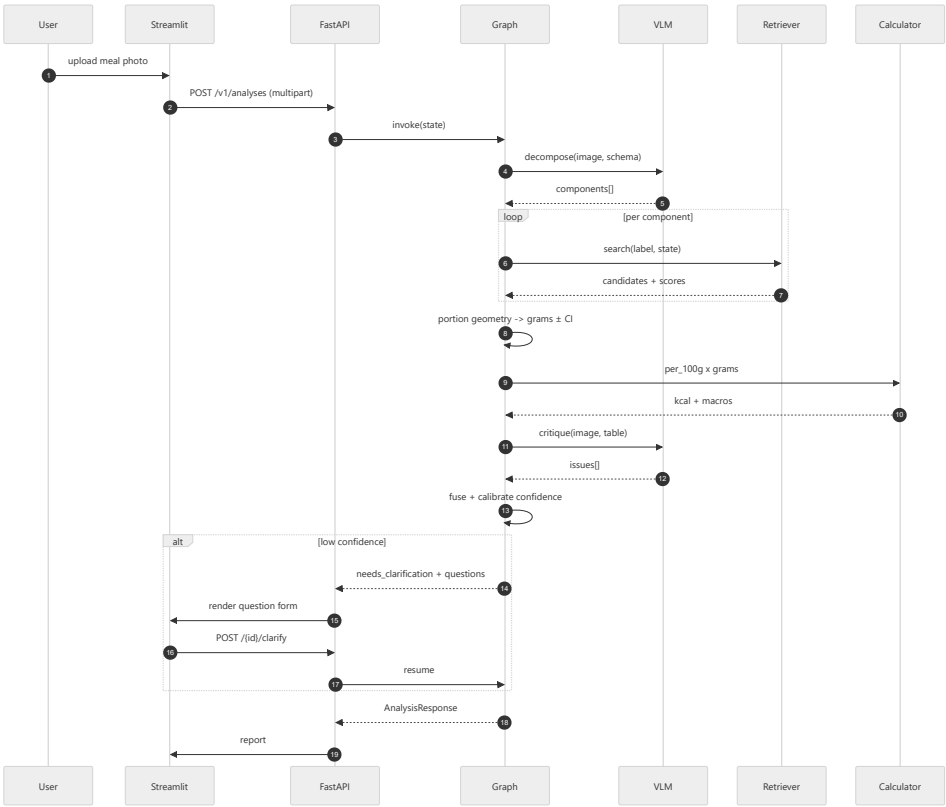
Error source	Countermeasure in this architecture
One dish label for a mixed plate	Component decomposition (M3) with bounding regions
Guessed serving size	Scale-anchored geometry → volume → mass, with an interval (M5)
Model-invented calories	Retrieval-grounded values + arithmetic outside the LLM (M6, M7)
Confidently wrong output	Fused, calibrated confidence + selective clarification (M8, M9)

3. Containers (C4 L2)



Why the core is framework-free: M1–M10 must be testable without a graph, a network, or a database. LangGraph wires them; it does not contain them. This is also what makes the 12-week plan survivable — if orchestration slips, the pipeline still ships.

4. End-to-end sequence



5. Confidence model

```
s_ident, s_retr, s_port, s_crit, s_img, s_state, s_cover   ∈ [0,1]

raw = 0.28·s_ident + 0.20·s_retr + 0.18·s_port
    + 0.14·s_crit  + 0.08·s_img  + 0.07·s_state + 0.05·s_cover

confidence = isotonic_calibrate(raw)        # fitted on dev split

Hard caps apply before calibration (`matched=False → ≤0.25`,
critic veto `→ ≤0.40`, severe blur `→ ≤0.55`).

Meal confidence = `coverage^0.5 × Σ w_i·c_i`, mass-weighted.
Totals get P10/P50/P90 by Monte-Carlo propagation of log-normal mass error
(fixed seed → reproducible reports).
```

Full derivation: `docs/06-confidence-and-clarification.md`.

6. Knowledge base

The system relies on a **dual-index retrieval** strategy that matches the two granularities of the data:

- **Ingredient-level** (IFCT 2017 + USDA snapshot) – the primary source of truth. Every food record is versioned (`kb_version`) and stored in SQLite; the vector index is derived from these rows, never written directly.
- **Composite-level** (dish recipes) – derived from ingredient records via recipe fractions computed by the same M7 arithmetic used at inference, preserving provenance at both granularities.

Related work (brief)

- **IFCT 2017** – Indian food composition tables (528 foods, ~150 nutrients per 100 g). Provides the ground-truth ingredient layer.
- **USDA FoodData Central** – public domain nutrient database; used as a fallback when the local KB is missing.
- **Dual-index retrieval** – hybrid BM25 + vector search (Chroma + BM25) with RRF fusion; this is the key data decision that enables both ingredient-level and composite-level lookups.
- **Adaptive routing** – T0/T1/T2 tiers based on quality, component count, and scene complexity; this is the core control plane.

7. Tech stack

What it does

An AI-powered food recognition and nutritional estimation system that analyzes meal images, identifies individual food components, estimates portion sizes, and calculates calories and macronutrients. It uses multimodal AI, hybrid retrieval, and confidence scoring to produce nutrition estimates, with clarification when predictions are uncertain.

Tech Stack

- **Language:** Python 3.11 / 3.12

- **Backend:** FastAPI, Uvicorn
- **Frontend:** Streamlit
- **AI Orchestration:** LangGraph
- **Vision Models:** Gemini 2.5 Flash (Groq and Ollama optional)
- **Embeddings:** BAAI/bge-small-en-v1.5
- **Reranking:** ms-marco-MiniLM-L-6-v2
- **Vector Database:** ChromaDB
- **Lexical Search:** BM25s
- **Database:** SQLite
- **Image Processing:** Pillow, OpenCV
- **Data Processing:** NumPy, Pandas, Scikit-learn
- **Validation:** Pydantic v2
- **Testing & Code Quality:** Pytest, Ruff, Mypy

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8. Interfaces

API (`/v1`)

Method	Path	Notes
POST	`/v1/analyses`	multipart image + optional context; returns `AnalysisResponse`
GET	`/v1/analyses/{id}`	replay a frozen report
POST	`/v1/analyses/{id}/clarify`	answers, resume graph, max 2 rounds
PATCH	`/v1/analyses/{id}/components/{ord}`	correct identity/mass, recompute, log feedback
POST	`/v1/analyses/{id}/components`	add a missed component
DELETE	`/v1/analyses/{id}`	delete analysis + artifacts

Method	Path	Notes
GET	`/v1/foods/search`	KB search for manual correction
GET	`/v1/meta`, `/healthz`	provider, KB version, dependency health

9. Non-functional targets

Dimension	Target
Latency T1	p50 ≤ 8 s, p95 ≤ 20 s
Cost	≤ $0.01 per analysis (T2)
Calorie error	MAPE ≤ 18 %, MAE ≤ 90 kcal on mixed plates
Component F1	≥ 0.80 at IoU 0.5
Calibration	ECE ≤ 0.10; selective accuracy ≥ 0.88 at conf ≥ 0.8
Offline	full pipeline with `MOCK_MODE=1`
Privacy	local-only images, 30-day retention, one-click delete

10. References (papers)

- **IFCT 2017** – Indian Food Composition Tables (NIN Hyderabad). Primary ingredient source.
- **USDA FoodData Central** – Public domain nutrient database; used as fallback when local KB is missing.
- **Dual-index retrieval** – Hybrid BM25 + vector search (Chroma + BM25) with RRF fusion; key design decision.
- **Adaptive routing** – T0/T1/T2 tiering based on quality, component count, and scene complexity; reduces unnecessary VLM calls.
- **Confidence calibration** – Isotonic calibration on dev split; ECE ≤ 0.10 target.
- **MOCK_MODE** – Deterministic fixtures keyed by image hash; enables offline reproducibility.